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REVIEW

Treatment of Primary Aldosteronism

Stéfanie Parisien-La Salle, Gregory L. Hundemer, Matthew A. Nehs, Justine A. Barletta, Anand Vaidya

ABSTRACT: Primary aldosteronism (PA) is a common cause of hypertension, characterized by renin-independent aldosterone production that drives inappropriate mineralocorticoid receptor activation, sodium retention, volume expansion, and potassium wasting, ultimately resulting in hypertension and adverse cardiorenal outcomes. Management of PA involves therapies that target these pathophysiologic mechanisms to restore homeostasis and reduce risk, which is usually tailored based on patient preference and whether PA is lateralizing or nonlateralizing. For patients with lateralizing PA, surgical adrenalectomy, and to a lesser extent, minimally invasive adrenal or adrenal artery ablation, is highly effective at improving blood pressure control and risk for incident cardiovascular outcomes. However, the vast majority of patients with PA will be treated with medical therapy using steroidal mineralocorticoid receptor antagonists as the cornerstone of therapy, and epithelial sodium channel inhibitors serving as infrequent alternatives. Dietary sodium restriction in PA should be strongly encouraged because it reduces the substrate that fuels PA pathophysiology; dietary sodium restriction can facilitate substantial reductions in blood pressure, especially when combined with mineralocorticoid receptor antagonist therapy. Once initiated, medical therapy should be intensified to achieve 3 objectives in the following order of importance: normalization of blood pressure with the fewest number of antihypertensive agents, normalization of serum potassium when applicable, and increases in renin from baseline as a biomarker of adequate aldosterone blockade.

Key Words: adrenalectomy ■ blood pressure ■ homeostasis ■ nephrons ■ potassium

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Primary aldosteronism (PA) is characterized by renin-independent aldosterone production that is inadequately suppressed by volume and sodium loading,[1,2] resulting in maladaptive MR (mineralocorticoid receptor) activation in both renal and extra-renal tissues[3] and incident cardiovascular and kidney injury.[3–10] In the distal nephron, MR activation induces sodium reabsorption via epithelial sodium channels (ENaC), resulting in extracellular volume expansion, glomerular hyperfiltration, physiological suppression of renin and angiotensin II, potassium wasting, and hypertension.[11] Thus, excessive sodium reabsorption plays a central pathogenic role in PA, with a low renin serving as a physiological reflection of the long-feedback loop induced by inappropriate aldosterone production.

PA is increasingly recognized as a common cause of hypertension, implicating the need for aldosterone-directed therapy in a growing population of patients.[12–21] Targeted therapy in PA has been shown to mitigate risk for incident cardiovascular disease,[22,23] by directly targeting key pathophysiologic mechanisms, including inappropriate MR activation, ENaC-mediated sodium retention, and aldosterone-producing adrenal micronodules or tumors.

Herein, we review established and emerging evidence supporting treatment strategies for PA, excluding aldosterone synthase inhibitors and nonsteroidal MR antagonists, which are reviewed in a separate article published in this issue.[24]

Clinical Case Vignette

A 38-year-old man was referred for management of hypertension and cardiomyopathy. He was diagnosed with hypertension at 28 years old and had multiple documented episodes of hypokalemia with serum potassium as low as 3.1 mEq/L. He was treated with lisinopril, hydrochlorothiazide, metoprolol, and potassium chloride, and his blood pressure was 140 to 150/80 to 100 mm Hg. ECG showed evidence of left ventricular hypertrophy, and cardiac magnetic resonance imaging showed hypertrophic cardiomyopathy and extensive late-gadolinium enhancement (indicative of myocardial fibrosis). He was diagnosed with PA after testing revealed a plasma

Correspondence to: Anand Vaidya, Center for Adrenal Disorders, Mass General Brigham, Harvard Medical School, Boston, MA 02115. Email anandvaidya@bwh.harvard.edu
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Nonstandard Abbreviations and Acronyms

ENaC	epithelial sodium channel
MR	mineralocorticoid receptor
PA	primary aldosteronism

aldosterone of 40 ng/dL, plasma renin activity of 0.33 ng/mL per hour, and an aldosterone-to-renin ratio of 121 ng/dL per ng/mL per hour. Given his young age, uncontrolled hypertension, and recurrent hypokalemia, and evidence of cardiovascular disease, he underwent genetic testing for inheritable forms of PA, which was unrevealing.

What are the key management considerations and therapeutic options for this young patient with PA and evidence of cardiovascular injury?

Nonpharmacological Treatments

Dietary Sodium Restriction

Sodium fuels PA pathogenesis. The vast majority of people in industrialized societies consume a very high sodium intake, typically 3 to 4 g of sodium per day, or more.[25,26] Studies of hunter-gatherer populations indicate that ancestral human sodium intake was likely substantially lower than current intake, probably below 50 mg/d.[27] Therefore, contemporary sodium intake is likely orders of magnitude higher than our homeostatic hormonal systems are designed to manage. When dietary sodium intake is this high, circulating and extracellular volume is relatively expanded, resulting in physiological suppression of the renin-angiotensin-aldosterone system. When aldosterone production persists, despite a high dietary sodium intake, volume expansion, and low renin levels, pathological aldosterone-MR interactions ensue, leading to progressive inflammation, fibrosis, and ultimately, cardiovascular and kidney injury.[28] High sodium intake and inappropriate aldosterone production act synergistically in driving cardiovascular remodeling in PA.[29–33] The magnitude of dietary sodium intake appears to influence adverse cardiac remodeling, as higher sodium intake is correlated with greater left ventricular hypertrophy and reductions in sodium intake after treatment are associated with a greater regression of cardiac mass in patients with PA, effects comparable to the effect of low-dose MRA (mineralocorticoid receptor antagonist).[29,30]

Salt is composed of sodium chloride, with 1 g of sodium corresponding to ≈2.5 g of salt.[26] Inappropriately high aldosterone production has been shown to alter sodium taste perception and increase salt appetite, resulting in a vicious cycle of increased salt intake in patients with PA (up to 10 g/d of salt).[34–36] Consequently, reducing sodium consumption in patients with PA not only lowers blood pressure, but if sufficiently restricted (eg, to <1000 mg/d), can also raise renin and lower the aldosterone-renin ratio, producing effects comparable to the use of MRAs.[37] Unfortunately, this degree of dietary sodium restriction is not sustainable for most patients. When combined with MRAs, moderate sodium restriction in patients with PA (reducing salt intake to ≈5 g/d) has been shown to significantly reduce systolic and diastolic blood pressure, pulse pressure, body weight, and normalize depressive symptoms.[38] As such, dietary sodium restriction is a critical component of PA therapy, yet is often overlooked or under-emphasized. All patients with hypertension are advised to restrict dietary sodium,[39], but patients with PA should be especially counseled on the critical and causal benefit of reducing sodium in their diet.

Weight Loss

Obesity is commonly regarded as a condition in which there is renin-angiotensin-aldosterone overactivation.[40] There is evidence that adipocyte-derived factors can stimulate aldosterone production.[41–43] Consistent with these pathophysiologic mechanisms, obesity commonly co-associates with hypertension, where the prevalence of PA is high, resulting in similarly high or higher rates of PA among individuals with obesity.[44] In a cohort of patients with obesity and early-stage hypertension, at least 37% of participants were found to have PA.[44] In another cohort of patients with resistant hypertension and obesity, 24-hour urinary aldosterone excretion was positively correlated with body mass index.[45] Interventions that induce weight loss are known to lower blood pressure[46] and may also favorably modulate aldosterone; therefore, encouraging weight loss is an intervention that should be encouraged for any patient with hypertension. In the SURMOUNT-1 BP trial, a study that investigated the effect of tirzepatide, a glucose-dependent insulinotropic polypeptide/ GLP1RA (glucagon-like peptide-1 receptor agonist), on 24-hour ambulatory blood pressure in individuals with body mass indexes ≥27 kg/m^2, blood pressure declined by 7.4 to 10.6 mmHg systolic, and this effect was partially independent of weight loss.[47] In a clinical trial of 285 participants, 4% to 7% weight loss over 24 months was associated with reductions in aldosterone levels, with lower aldosterone correlating with favorable changes in obesity-related factors, including reduced C-reactive protein, leptin, insulin resistance, and increased adiponectin.[48] Moreover, in patients with diabetes and advanced kidney disease, a single dose of liraglutide reduced plasma aldosterone concentrations compared with placebo, with increased natriuresis and elevated atrial natriuretic peptide, suggesting a potential suppression of aldosterone action.[49] Similarly, weight loss increases B-type natriuretic peptide, which suppresses aldosterone through multiple mechanisms, including angiotensin II-independent pathways, potentially benefiting hyperaldosteronism.[50–54] Together, these findings provide early evidence that weight loss and GLP1RA

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may favorably modulate aldosterone and that dedicated studies should investigate outcomes in patients with PA.

Pharmacological and Interventional Treatments

The introduction of CYP11B2 immunohistochemistry has shifted the traditional dichotomy of unilateral versus bilateral PA towards a more nuanced and physiological understanding of PA pathogenesis and evolution.[55] We now understand that pathogenic somatic mutations underlie the vast majority of PA cases, often manifesting across a spectrum of morphologies visualized on CYP11B2-mediated immunohistochemistry, from aldosterone-producing micronodules to aldosterone-producing adenoma.[55] Moreover, the observation that CYP11B2-positive micronodules may accumulate with age[56–58] has further reinforced the notion that PA is predominantly a condition that emanates from diffuse foci of aldosterone production that can arise from both adrenal glands (at times asymmetrically or asynchronously), as observed in at least two-thirds of adrenal vein sampling cases.[59] The predominance of PA affecting both adrenal glands is likely underestimated, as mild-to-moderate, or subclinical, forms of PA are rarely diagnosed, but are usually nonlateralizing,[60] and even cases of PA that lateralize may have aldosterone-producing foci in the contralateral gland that contribute to persistent and recurrent PA.[61,62] Although unilateral adrenalectomy for PA can induce short-term biochemical remission, long-term recurrence is more common than traditionally realized, particularly in patients with multiple aldosterone-producing nodules or micronodules, or aldosterone-producing diffuse hyperplasia on pathology.[62] Thus, the traditional dichotomy between pharmacological and interventional therapy has increasingly been blurred, giving way to a treatment paradigm that includes overlapping approaches (Figure 1).

Interventional Treatments

Interventional treatments can be considered in suitable candidates, particularly when they have a PA that lateralizes to 1 adrenal gland.

Surgical Adrenalectomy

Laparoscopic surgical adrenalectomy remains the gold standard treatment for patients with lateralizing PA (Figure 2A). Open adrenalectomies were replaced with the less invasive laparoscopic technique starting in the early 1990s.[63,64] Minimally invasive adrenalectomy can be performed using different methods, including transperitoneal, posterior retroperitoneoscopic, hand-assisted, and robot-assisted techniques. The choice of operation depends on patient anatomy, tumor characteristics, prior abdominal operations, and surgeon expertise.[65] Reported complication rates for laparoscopic adrenalectomy are relatively low, with morbidity rates ranging from 5% to 14% and mortality <1%.[66,67] However, specifically for PA, complication rates following adrenalectomy appear to be lower, averaging ≈4.7%.[68] Uncommon major complications include bleeding requiring transfusion or conversion to open surgery, pneumothorax, pneumonia, and myocardial infarction. Rare and mild complications include temporary abdominal wall numbness.[66] Adrenal-sparing surgery is not typically considered in PA because histopathologic studies often demonstrate multifocal aldosterone-producing lesions that risk leaving persistent aldosterone production postoperatively.[69] Unilateral adrenalectomy has also shown clinical benefits in patients with nonlateralizing or bilateral PA, particularly when there is asymmetrical lateralization favoring 1 side and asymmetrical adrenal morphology.[70–75]

There is substantial evidence that unilateral adrenalectomy for lateralizing PA provides robust blood pressure control and reductions in adverse cardiovascular outcomes compared with medical therapy.[23,76–81] In a 2024 meta-analysis that included over 15 000 patients with PA, surgery was associated with an overall lower risk of death, major adverse cardiovascular events, hospitalization for heart failure, and progression to chronic kidney disease and nonsignificant reductions in myocardial infarction and stroke compared with medical therapy.[76] Another recent systematic review supporting the updated 2025 Endocrine Society guidelines included nearly 18 000 patients with PA and found that, compared with surgical therapy, PA-specific medical therapy was associated with higher systolic blood pressure levels, a greater number and higher doses of antihypertensive medications, and increased risks of stroke, heart failure, and all-cause mortality.[77] No statistically significant differences were observed between treatments for ischemic heart disease, atrial fibrillation, hypertension remission, major adverse cardiovascular events, or cardiovascular mortality.[77] These studies should be interpreted in the context of their limitations, including suboptimal MRA dosing and adherence, variability in disease severity among groups, and selection bias.[6,82]

Finally, surgical adrenalectomy is also associated with more favorable cardiac remodeling (greater regression of left ventricular mass and a lower risk of persistent hypertrophy) compared with medical therapy.[80] A recent meta-analysis showed that both adrenalectomy and MRAs improve left ventricular remodeling in PA,[82] with adrenalectomy being associated with slightly greater short-term left ventricular mass index reduction, while long-term reductions were similar overall. However, in unilateral PA, adrenalectomy resulted in significantly greater left ventricular mass index improvement than medical therapy.[82]

Adrenal Ablation and Adrenal Artery Ablation

Adrenal ablation is a commonly available minimally invasive treatment option for patients with lateralized PA

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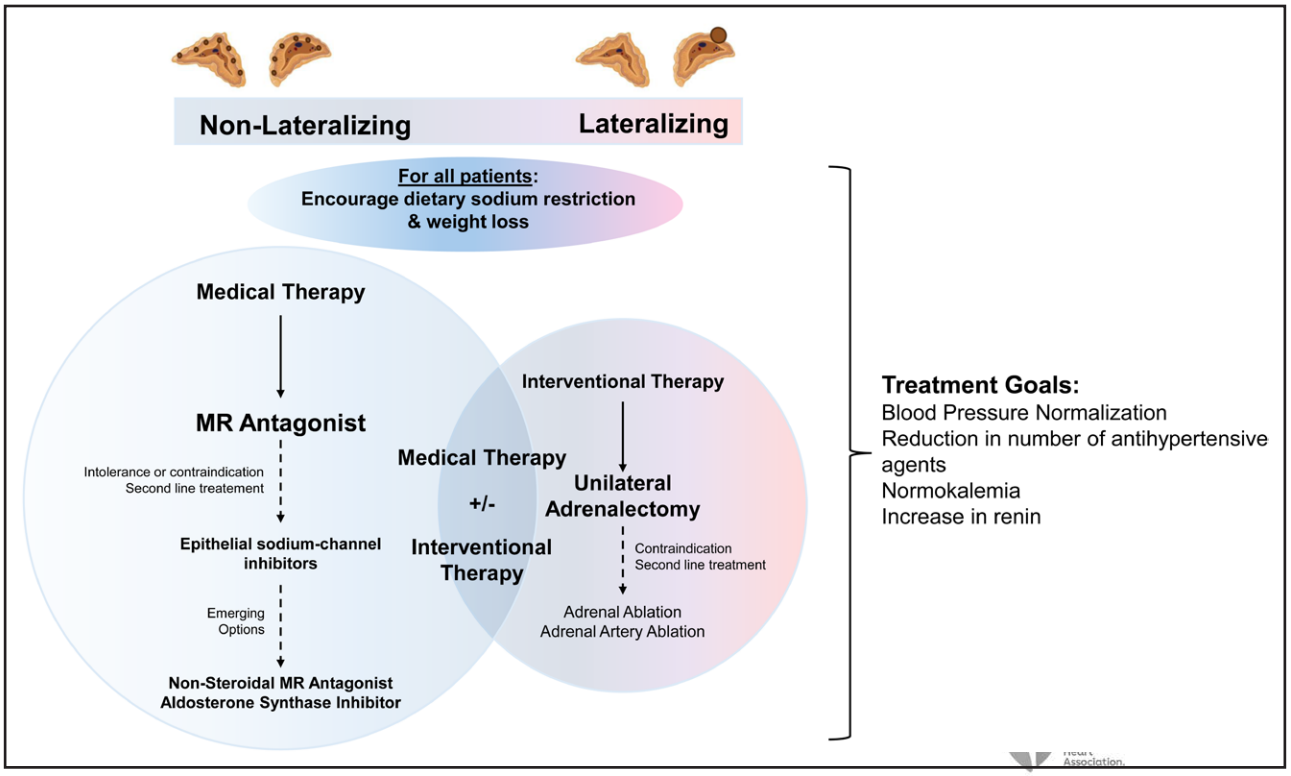

Figure 1. Therapeutic framework for the management of primary aldosteronism (PA).
Management strategies for PA are guided by disease lateralization and patient-specific factors. Lifestyle interventions such as dietary sodium restriction and weight loss (particularly for those who are overweight or obese) are recommended for all patients with hypertension. For patients with nonlateralizing PA, those with lateralizing PA who do not undergo interventional therapy, and those with persistent or recurrent PA following interventional therapy with bilateral disease, MR (mineralocorticoid receptor) antagonists are the first line of treatment. In cases of intolerance or contraindication, epithelial sodium channel inhibitors may be used, with emerging therapy, including nonsteroidal MR antagonists and aldosterone synthase inhibitors, under investigation. In lateralized disease, interventional therapy with unilateral adrenalectomy remains the preferred approach, with adrenal or adrenal artery ablation considered in selected patients who are not surgical candidates. Medical and interventional strategies may overlap depending on treatment response and disease course. Across treatment modalities, shared goals include blood pressure normalization, correction of hypokalemia, reduction in antihypertensive medication burden, and suppression of renin.

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who are not candidates for surgery or prefer alternatives to surgery, using techniques such as CT-guided percutaneous radiofrequency ablation (RFA), endoscopic ultrasound-guided RFA, and catheter-based adrenal artery ablation[83–90] (Figure 2B). RFA has typically been used when adrenal nodules are visible on imaging, whereas adrenal artery ablation may have utility regardless of whether a nodule is visible or not. CT-guided RFA is the most commonly used technique. It requires placement of ablation probes directly into adrenal cortical tissue (usually an adenoma or nodule) using real-time imaging, followed by delivery of thermal energy to induce tissue destruction. The procedure is typically performed under general anesthesia.[83,91] Possible adverse events from RFA include pain, pneumothorax, hemorrhage, injury or perforation of nearby organs, and hypertensive crises due to catecholamine release.[83,92] Incomplete ablation may leave residual adrenocortical tissue harboring aldosterone-producing micronodules, thereby predisposing to persistent or recurrent PA.

A novel technique for left-sided aldosterone-producing adenomas is endoscopic ultrasound-guided RFA, which consists of ablating aldosterone-producing adenomas through the stomach.[90] This technique was evaluated in the FABULAS proof-of-concept study (n=28), where aldosterone-producing adenomas identified by adrenal vein sampling or molecular nuclear imaging ([^{11} C] MTO or [^{18} F] CETO PET-CT) were successfully ablated without major complications.[90] In this novel study, adrenal tissue was obtained by endoscopic ultrasound-guided fine-needle biopsy, enabling assessment of CYP11B2 expression.[90] However, unlike whole adrenalectomy specimens, this approach was unable to comprehensively surrounding adrenal cortex for additional CYP11B2-positive micronodules, which may help inform the risk of persistent or recurrent disease. Three months following the procedure, 75% of participants had biochemical remission, while 43% had complete or partial improvement in their hypertension. In 4 patients, complete lesion disappearance was accompanied by normal blood pressure without the use of medications.[90]

Catheter-based adrenal artery ablation is another technique shown to be safe and efficacious in the treatment

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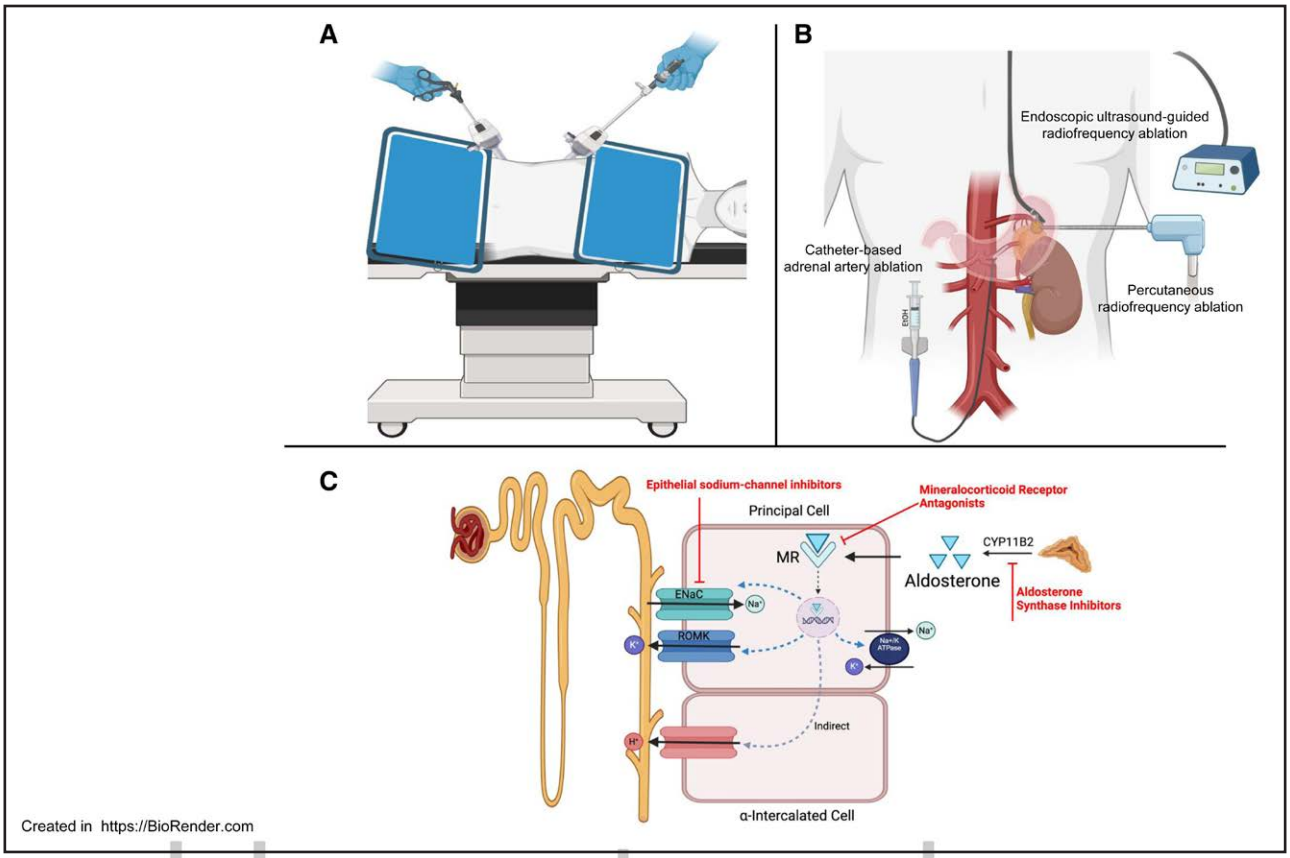

Figure 2. Interventional and medical treatments for primary aldosteronism (PA).
A, Laparoscopic unilateral adrenalectomy: the standard treatment for lateralized PA. **B**, Adrenal and adrenal artery ablations: Catheter-based adrenal artery ablation and percutaneous or endoscopic ultrasound-guided radiofrequency ablation are used in selected patients with lateralized PA who are not candidates or refuse surgery. **C**, Medical treatment mechanisms and targets: aldosterone binds the MR (mineralocorticoid receptor) in principal cells of the distal nephron, promoting epithelial sodium channel (ENaC) mediated sodium reabsorption and potassium secretion via ROMK, with indirect effects on hydrogen ion handling in intercalated cells. Therapeutic strategies include MR antagonists, ENaC inhibitors, and aldosterone synthase inhibitors, which reduce aldosterone production by inhibiting CYP11B2. Figures created using Biorender.com.

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of lateralizing PA, including in cases of PA without a clear adrenal nodule.[85,88] Adrenal artery ablation consists of infusing ethanol into the adrenal artery via a microcatheter, a procedure usually done under local anesthesia.[83] In a randomized controlled trial of patients with lateralizing PA, catheter-based adrenal ablation produced a greater reduction in office systolic blood pressure (−18 mm Hg) at 1 month than spironolactone therapy, without any major complications from the procedure.[85] This reduction in blood pressure was no longer evident at the 6-month follow-up; however, the ablation group remained with a significant reduction in the defined daily dose of home antihypertensive medications.[85] One potential explanation for this lack of durability in blood pressure lowering may have been the presence of aldosterone-producing micronodules in the contralateral adrenal gland; given the absence of pathology resulting from this procedure, it was not possible to further evaluate this hypothesis. In another study with 24 months of follow-up and including 109 patients with PA, unilateral adrenal artery embolization demonstrated reductions in 24-hour blood pressure (−9.9/−6.4 mm Hg), complete and partial improvements in hypertension in 14.9% and 50.0% of patients, complete and partial biochemical success in 13.6% and 45.5%, and improved cardiac remodeling with no major long-term safety concerns.[93]

Overall, adrenal tissue and artery ablation appear to perform as well or better than medical therapy, but may not be as effective as surgery.[84] A recent meta-analysis that included over 10 000 individuals from 47 studies compared adrenalectomy versus adrenal ablation (radiofrequency, catheter-based chemical ablation, and embolization) versus MRA therapy for PA.[84] The primary outcomes were changes in blood pressure, serum potassium correction, and improvements in the aldosterone-renin ratio. Adrenalectomy outperformed the other treatments, followed by ablation procedures, then MRA therapy. Notably, these results were influenced by obligate selection bias inherent to PA subtype and treatment choice (ie, MRA therapy is typically reserved for nonlateralizing PA, whereas adrenalectomy and ablation are typically offered for lateralizing PA). Nevertheless, ablation

may serve as a reasonable alternative when surgery is not feasible or desired.[84] Limitations include paucity of long-term efficacy data and the absence of a whole pathological specimen.

Pharmacological Treatments

Before initiating medical therapy for PA, nonpharmacological approaches (dietary sodium restriction and weight loss) should be strongly advised. In conjunction, PA-specific medical therapy (Table 1) should be initiated in patients with nonlateralizing PA, those with lateralizing PA who do not undergo interventional therapy, and those with persistent or recurrent PA following interventional therapy.

Steroidal Mineralocorticoid Receptor Antagonists

Steroidal MRAs are the standard medical therapy for PA.[60] Their cost, availability, and adverse event profiles are key considerations when selecting an MRA (Table 1; Figure 2C).

Spironolactone

The 2025 Endocrine Society guidelines advise using spironolactone as the preferred MRA option due to its affordability and widespread availability.[60] Spironolactone emerged in the 1950s as a potent antagonist of aldosterone's sodium-retaining and potassium-wasting actions.[94,95] Spironolactone is a progesterone-derived, nonselective steroidal antagonist that competitively blocks the MR, while also binding androgen and progesterone receptors.[96] These off-target interactions account for its anti-androgenic adverse-effect profile (gynecomastia and low libido in men, and breast tenderness and menstrual irregularities in women), which can limit tolerability and should be discussed with patients before initiation.[60] Gynecomastia rates in men taking spironolactone vary from 10% to nearly 30%, depending on studies and dosing.[97,98] The starting dose is typically around 12.5 to 25 mg daily, with titration following a step-wise approach.[60] The antiandrogenic effects in men increase with longer treatment duration and higher doses, becoming more apparent at doses ≥50 mg daily in a dose-dependent manner.[99]

Eplerenone

Eplerenone is a steroidal MRA developed in 1987.[100] Incorporating a 9α,11α-epoxy group preserves MR antagonism while markedly reducing affinity for the androgen and progesterone receptors.[100] Although eplerenone is roughly half as potent as spironolactone, it carries a minimal risk of antiandrogenic side effects, making it the preferred choice in men.[101,102] The starting dose is usually around 25 mg twice daily, as eplerenone's shorter half-life requires divided dosing to maintain adequate MR blockade.[103]

MRAs have been shown to be effective at lowering blood pressure in the treatment of PA[104] and resistant hypertension.[105] When compared, spironolactone lowered blood pressure more effectively than eplerenone in patients with PA, although it caused more side effects.[106]

The scope of MRA treatment extends beyond blood pressure and potassium management by mitigating end-organ damage.[6,23,107–112] However, this appears to be conditional on adequate titration of MRA treatment (discussed later).[23]

Epithelial Sodium-Channel Inhibitors

The 2025 Endocrine Society guidelines recommend ENaC inhibitors (amiloride, triamterene) as an alternative when MRAs are not tolerated or available (Table 1; Figure 3C).[60] By blocking sodium reabsorption in the

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Table 1. XXX

Medical therapy	Starting dose	Maximum dose	Adverse effects	Monitoring
Steroidal mineralocorticoid receptor antagonist				
Spironolactone	12.5–25 mg daily	200 mg daily*	Hyperkalemia, antiandrogenic effects (gynecomastia, sexual dysfunction, menstrual irregularities)	Serum potassium, renal function, blood pressure, renin; monitor at 2–3 mo and after dose changes; more frequent in renal impairment or severe hypokalemia
Eplerenone	25 mg twice daily	200 mg twice daily*	Hyperkalemia, less antiandrogenic effects than spironolactone	
Nonsteroidal mineralocorticoid receptor antagonist				
Finerenone	10 mg daily	20 mg daily	Hyperkalemia, less antiandrogenic effects than spironolactone	Serum potassium, renal function, blood pressure, renin; monitor at 2–3 mo and after dose changes; more frequent in renal impairment or severe hypokalemia
Esaxerenone	2.5 mg daily	5 mg daily	Hyperkalemia, less antiandrogenic effects than spironolactone	
Epithelial sodium-channel inhibitors				
Amiloride	2.5–5 mg daily	40 mg daily	Hyperkalemia, nausea, dizziness	Serum potassium, renal function, blood pressure, renin; monitor at 2–3 mo and after dose changes; more frequent in renal impairment or severe hypokalemia
Triamterene	37.5–50 mg daily	300 mg daily	Hyperkalemia, nephrolithiasis, nausea	

*Doses >200 mg daily have been safely given in PA.

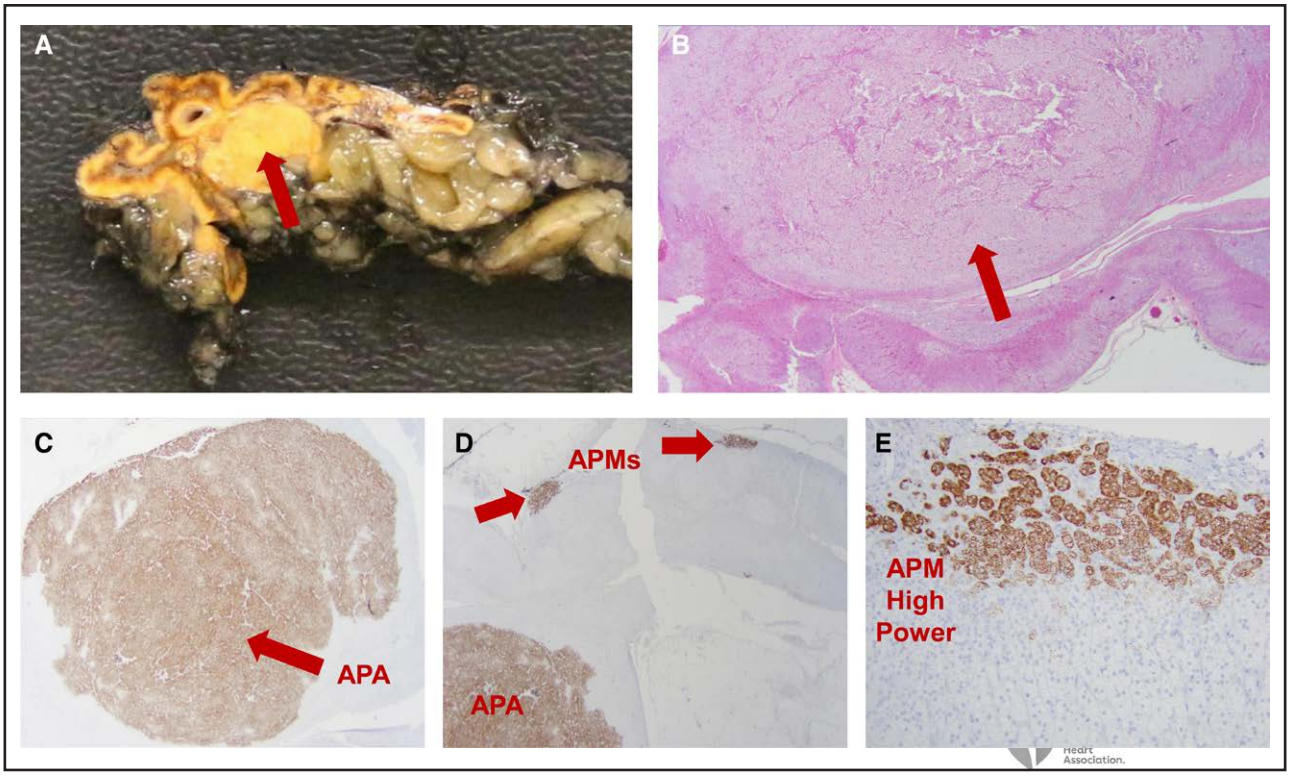

Figure 3. Surgical pathology.
A, Gross pathology revealing a normal appearing right adrenal gland with a 1.0 cm adrenal cortical adenoma emanating from the body of the adrenal gland (red arrow). **B**, Hematoxylin and eosin-stained section of the adrenal gland demonstrating the adrenal cortical adenoma (red arrow) and surrounding adrenal cortex. **C**, Immunohistochemistry for CYP11B2 demonstrating positive staining within the adenoma, thereby confirming it to be an aldosterone-producing adenoma (APA; red arrow). **D**, Two CYP11B2-positive aldosterone-producing micronodules (APM) seen adjacent to the aldosterone-producing adenoma in the normal cortex (red arrows). **E**, High-power magnification of 1 CYP11B2-positive APM.

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distal nephron, ENaC inhibitors can lower blood pressure and treat hypokalemia; however, they do not prevent MR-driven tissue damage. Amiloride reversibly inhibits ENaC in the distal nephron and has been used to treat PA since the 1980s, demonstrating effectiveness in lowering blood pressure and correcting hypokalemia.[113,114] The starting dose can range from 2.5 to 10 mg daily. Triamterene is less potent than amiloride and may carry a higher risk of kidney stones.[115]

There are no studies that directly compare MRAs versus ENaCs for the treatment of PA.[60] However, in studies comparing both agents in resistant hypertension, ENaCs may be noninferior to MRAs in regard to blood pressure lowering, but their impact on reducing MR-mediated organ damage remains unknown.[60,105,116]

Emerging Therapies

Non-steroidal MRAs have shown cardiovascular and renal benefits in patients with diabetes and CKD or heart failure,[117–119] although growing evidence in PA remains limited.[120,121] These agents have a much lower affinity for the androgen or progesterone receptors, resulting in a more favorable side-effect profile.[122] ASIs (aldosterone synthase inhibitors) are a new class of drugs that act by blocking CYP11B2, the enzyme responsible for aldosterone synthesis. Phase 2 and phase 3 studies have demonstrated important reductions in blood pressure in patients with resistant and uncontrolled hypertension.[123–126] A small phase 2a open-label study showed that baxdrostat substantially reduced systolic blood pressure (25 mm Hg), aldosterone, and aldosterone-renin ratio, while correcting hypokalemia in patients with PA.[127] These therapies will be addressed in a separate review published in this issue.[24]

Treatment Objectives During Medical Treatment for PA

Once medical therapy is initiated, the primary objectives are normalization of blood pressure and serum potassium.[60] Measuring renin can serve as a guide to optimize MRA dose titration.[128] Improving overall quality of life is also an important goal of treatment. Table 2 outlines a streamlined algorithm for optimizing medical treatment in PA based on blood pressure, potassium, and renin status.

Blood Pressure Control

The American Heart Association and American College of Cardiology guidelines classify systolic pressures of

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Blood pressure	K^+ status	Renin status relative to baseline	Recommended action
High	Normal or low	Lower/same	Increase MRA dose
	Normal	Higher	Increase MRA, OR
			Increase non-MRA antihypertensive agents
	Low	Higher	Increase MRA dose
	High	Any	Add a thiazide or loop diuretic, SGLT2 inhibitor, or novel potassium binder. If no improvement in potassium, consider lowering the MRA dose and increasing the non-MRA antihypertensive agents.
At target	Normal	Lower/same	No change, OR
			Reduce non-MRA antihypertensive agents to allow further MRA up-titration, if applicable.
	Low	Lower/same	Reduce non-MRA antihypertensive agents to allow further MRA up-titration, if applicable.
	Normal	Higher	Continue current regimen
	High	Any	Stop ACE inhibitors or angiotensin receptor blockers and consider potassium binder therapy.

MRA indicates mineralocorticoid receptor antagonist; and OR, odds ratio.

120 to 129 mm Hg with diastolic <80 mm Hg as elevated, 130 to 139 mm Hg, or 80 to 89 mm Hg as stage 1 hypertension and ≥140 mm Hg or ≥90 mm Hg as stage 2 hypertension.[39] There is very strong evidence that lowering blood pressure significantly reduces risk for adverse cardiovascular events.[129,130] Thus, for adults with confirmed hypertension, the 2025 American Heart Association/American College of Cardiology guidelines[39] recommend targeting a blood pressure below 130/80 mm Hg, with additional encouragement to achieve a systolic pressure under 120 mm Hg when possible, following evidence from the SPRINT, STEP, and BPROAD trial.[131–133] These blood pressure targets are derived from general hypertension populations rather than PA-specific cohorts. Even with adequate aldosterone-directed therapy and biomarkers consistent with appropriate MR blockade, patients may remain with blood pressures above 120/80 mm Hg and require additional non-PA antihypertensive therapy.

Across observational studies, MRAs have been shown to achieve systolic blood pressure reductions of up to about 25% of people, with nearly half of patients reaching adequate blood pressure control while needing fewer antihypertensive medications than before treatment.[104,107,134] However, there exist barriers to optimal blood pressure management. Firstly, blood pressure measurement variability is substantial. In a real-world cohort of over half a million adults with >7.7 million blood pressure readings, the average visit-to-visit systolic variability was 10.6 mm Hg, highlighting the limitations of relying solely on episodic in-clinic measurements.[135] Accurate blood pressure assessment, both in the clinic and at home, requires using a validated device with the correct cuff size, ensuring proper patient preparation (no smoking, caffeine, or exercise 30 minutes prior) and positioning (arm supported at heart level), allowing several minutes of rest, and obtaining multiple readings that are averaged to guide management.[39]

Second, reliance on blood pressure control alone seems to be insufficient for patients with PA.[23] Independent of blood pressure levels, patients with PA exhibit a higher risk of adverse cardiovascular events, target-organ injury, and both incident and progressive chronic kidney disease compared with individuals with essential hypertension.[4,9,23,136,137] This was further demonstrated in a large cohort study showing that patients with PA treated with MRAs had almost twice the rate of cardiovascular events compared with matched patients with essential hypertension (56.3 versus 26.6 events per 1000 person-years; hazard ratio [HR], 1.91), despite having similar blood pressure levels.[23] They also had higher risks of mortality (HR, 1.34), atrial fibrillation (HR, 1.93), and diabetes (HR, 1.26).[23] Thus, normalization of blood pressure is a critical priority of PA treatment, but is not the only therapeutic objective.

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Potassium Normalization Without Supplementation

Hypokalemia is reported in ≈9% to 37% of patients with PA, with higher rates in those harboring aldosterone-producing adenomas, generally reflecting a more severe form of PA.[138] Profound hypokalemia is relatively rare, but its presence is highly suggestive of PA.[139] Hypokalemia is associated with higher cardiovascular morbidity in patients with PA.[140,141] It is uncertain if this excess morbidity is a reflection of a more severe form of PA, the hypokalemia itself, or if both factors play an independent role. Thus, potassium normalization is a cornerstone of PA management and is included as a biochemical criterion in both the Primary Aldosteronism Surgical Outcome and the Primary Aldosteronism Medical Outcome criteria.[142,143]

Both MRAs and ENaCs have been shown to normalize potassium in PA treatment with adequate dosing.[144] There are no specific target levels for potassium in PA;

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however, studies from chronic kidney disease and heart failure populations suggest that there is a U-shaped association between serum potassium levels and mortality.[145–147] In a large meta-analysis of over 227 000 patients, both hypokalemia (RR, 1.35) and hyperkalemia (RR, 1.33) were associated with increased long-term all-cause mortality, with the lowest mortality observed at a potassium level of ≈4.2 mmol/L.[147] However, in PA, achieving biochemical success does not necessarily translate into normalizing blood pressure or mitigating cardiovascular risk.[142,143,148]

Optimizing Renin

Normalizing blood pressure and potassium may not fully mitigate cardiovascular risk in patients with PA.[23] Growing evidence from retrospective observational studies shows that PA should not be treated solely as a hypertensive disorder, but as an endocrine disorder, where a biomarker of appropriate MR blockade may be an increase in renin.[23,111,112,149] An increase in renin during aldosterone-targeted therapy reflects the reversal of the underlying PA pathophysiology, potentially indicating sufficient MR blockade to contract plasma volume, resolve glomerular hyperfiltration, and, in turn, release renin from its previously suppressed state.

A recent systematic review and meta-analysis supporting the new Endocrine Society guidelines[77] concluded that a low or suppressed renin was associated with an increased number of antihypertensive agents,[150,151] an increased risk of stroke,[152] an increased risk of atrial fibrillation,[110] and possibly an increased risk of cardiovascular mortality,[23] compared with patients with PA without a low or suppressed renin. Although this systematic review did not find significant differences in major cardiovascular events,[77] individual retrospective cohort studies did demonstrate a significantly lower risk for major cardiovascular events in patients with PA whose renin increased in response to MRA therapy.[6,23] Another systematic review and meta-analysis that addressed the same question and analyzed the same studies with slightly different inclusion and exclusion criteria concluded that patients with PA who achieved a renin activity>1.0 ng/mL/h during MRA therapy had a lower risk of incident cardiovascular events compared with those with persistently low renin (HR, 0.43 [95% CI, 0.23–0.80]).[149] Low renin during MRA therapy has also been associated with adverse cardiac remodeling,[109,150,153] and possibly with worse kidney outcomes[111,137,149]

In light of this evidence, the 2025 Endocrine Society Guidelines recommend that in patients with PA treated medically who have high blood pressure and low renin, intensification of PA-targeted medical therapy should be pursued with the goal of increasing renin levels from baseline.[60] If blood pressure normalizes, but renin remains low, other non-MRA anti-hypertensive agents can be reduced or discontinued to allow further intensification of the MRA dose with the goal of raising renin.[60] The Endocrine Society guidelines minimize emphasis on a specific renin target to achieve, as limited to no evidence supports a specific threshold. In contrast, an expert consensus statement published in 2025 proposed defining complete biochemical response to MRA therapy as a plasma renin activity >1.0 ng/mL per hour or renin concentration >10 mU/L, whereas not achieving these thresholds was defined as partial and absent responses.[143] Finally, in patients treated only with MRAs whose blood pressure is already at goal, the potential benefits of further increasing the dose to raise renin remain uncertain.[60] Prospective studies to evaluate renin-targeted therapy are needed, and some are currently underway (URL: https://www.clinicaltrials.gov; Unique identifier: NCT06108427).

Improving Quality of Life

Untreated PA is associated with a reduction in quality of life, impacting both physical and psychological well-being.[154–157] In a post hoc analysis of the SPARTACUS trial, both surgical and medical treatment improved quality of life, with surgical treatment producing greater improvements.[158] In a 2024 prospective cohort of 116 patients with PA, improvement in health-related quality of life after standardized therapy was strongly associated with correction of hypokalemia and effective blood pressure control.[159] Together, these findings highlight that quality of life is a meaningful, patient-centered indicator and should be incorporated into decision-making for PA.

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BARRIERS TO EFFECTIVE MEDICAL TREATMENT

Estimated Glomerular Filtration Rate Decline

PA is characterized by glomerular hyperfiltration driven by chronic volume expansion and increased renal plasma flow.[136,160–162] Initiation of aldosterone-directed therapy can reverse this state, almost always resulting in a decline in eGFR.[136,160,161] This initial drop reflects the correction of hyperfiltration rather than new kidney injury, and subsequent long-term eGFR loss tends to slow or stabilize after treatment.[163–165] This benefit seems to extend beyond blood pressure control and likely reflects diminished mineralocorticoid receptor activation within the kidney.[164] Overall, aldosterone-targeted therapy reverses the maladaptive renal physiology of PA and promotes long-term preservation of kidney function.

Hyperkalemia

In the context that potassium is usually normal or on the lower side, hyperkalemia resulting from MRAs or ENaCs

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is uncommon in PA. However, the risk of hyperkalemia increases with older age (>65 years), chronic kidney disease, and diabetes.[166] If hyperkalemia occurs, management strategies include stopping ACE inhibitors or angiotensin receptor blockers and, if applicable, adding a loop diuretic, thiazide, SGLT2 inhibitor, or novel potassium binder.[112,166]

Antiandrogen Effects

The antiandrogenic side effects associated with spironolactone (described above) can lead to discontinuation of MRA therapy, particularly in men.[167] When side-effects become important, patients can be switched to a more specific steroidal MRA (eplerenone) or an ENaC inhibitor. Newer therapies, including nonsteroidal MRAs and aldosterone synthase inhibitors, do not cause antiandrogenic effects and therefore, offer promising alternatives for patients who are unable to tolerate spironolactone.

Clinical Case Vignette Continued

Once diagnosed with PA, the patient was advised to strongly consider cross-sectional abdominal imaging and adrenal venous sampling to determine whether he might be a candidate for a unilateral adrenalectomy, mainly because of his young age and evidence of cardiovascular disease. He was initially averse to considering any interventional procedures and instead strongly favored a noninvasive medical treatment approach. Thus, nonpharmacological and pharmacological therapy were initiated. The patient was advised to reduce dietary sodium intake to below 1500 mg per day and was started on eplerenone 25 mg twice daily in addition to his current medications. Over the next 5 months, the dose of eplerenone was intensified, while his other medication doses and potassium supplements were de-escalated, with simultaneous monitoring of his home blood pressure and laboratory values. Five months later, while on eplerenone 75 mg twice daily and hydrochlorothiazide 25 mg daily, his blood pressure was normal (≈120/70 mm Hg), serum potassium was 4.3 mEq/L, and plasma renin activity was 7.3 ng/mL per hour. In this regard, his medical therapy had been optimized by intensifying MRA dosing to normalize blood pressure, minimize polypharmacy, normalize serum potassium, and induce a rise in renin as a biomarker of adequate aldosterone blockade.

Six months after initiating eplerenone, the patient was amenable to considering eligibility for unilateral adrenalectomy. Abdominal computed tomography revealed a possible subcentimeter right adrenal nodule. Eplerenone was stopped for 2 weeks, after which he underwent adrenal venous sampling, which revealed lateralization to the right adrenal. He underwent uncomplicated right laparoscopic adrenalectomy, and pathology revealed a 1.0 cm right adrenal cortical adenoma that was CYP11B2 positive with multiple aldosterone-producing CYP11B2-positive micronodules in the adjacent adrenal cortex (Figure 3). One month after the surgery, he was on no blood pressure medications or potassium supplements, blood pressure was 138/82 mm Hg, plasma aldosterone was 5 ng/dL, plasma renin activity was 2.5 ng/mL/h, and serum potassium was 4.3 mEq/L. He was advised to start amlodipine monotherapy for blood pressure control. He was counseled that his PA was in remission and that his cardiovascular risk profile was deemed to be improved. He was advised to continue a low-sodium diet and to return for annual follow-up to monitor for worsening hypertension and potential recurrence of PA, given the pathology showing multiple aldosterone-producing micronodules.

CLINICAL CASE CONCLUSION

This case highlights the different management pathways for PA, incorporating patient preference with reassessment over time. The case emphasizes the need for overlapping nonpharmacological, pharmacological, and interventional treatments, the approach to optimizing medical therapy and intensifying mineralocorticoid receptor antagonist dosing, and the outcomes of surgical adrenalectomy for lateralizing disease. Finally, it underscores the prognostic importance of adrenal histopathology in informing recurrence risk and the determination of long-term follow-up and patient counseling.

CONCLUSIONS

PA is a prevalent and treatable cause of hypertension. Medical and interventional treatments can target the underlying pathophysiology of PA; however, optimal outcomes depend on the ability to assess treatment adequacy. As the number of patients diagnosed with PA grows, reliable biomarkers to guide therapy selection, titration, and monitoring are essential to individualize care and mitigate long-term risk. The vast majority of patients with PA will be treated with MRAs, clinicians should be well-versed on their initiation, intensification, and the objectives for optimal therapy. However, in the near future, it is likely that novel medical and non-pharmacological therapies will emerge and disrupt this paradigm.

ARTICLE INFORMATION

Affiliations

Division of Endocrinology, Diabetes, and Metabolism, Center for Adrenal Disorders (S.P.-L.S., A.V.), Department of Surgery (M.A.N.), and Department of Pathology (J.A.B.), Mass General Brigham, Harvard Medical School, Boston, MA. Division of Endocrinology, Department of Medicine, Centre Hospitalier de l'Université de Montréal (CHUM), Université de Montréal, QC, Canada (S.P.-L.S.). Division of Nephrology, Department of Medicine, University of Ottawa, ON, Canada (G.L.H.).

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